

02LQD Specification and Simulation of Digital Systems

2022/01/26

1. Combinational circuit design (2 points)

Design a combinational circuit that receives a 4-bit unsigned binary number $X = x_3x_2x_1x_0$ and outputs a 4-bit unsigned binary number $Y = y_3y_2y_1y_0$ where $Y = (X + 5) \bmod 16$. Design in VHDL a combinational circuit implementing the above specification according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (12 points)

A serial combination lock allows entry to a locked room. It works as follows:

1. the user inputs the first combination and presses the ENTER button. The combination is recognized by a combinational network (whose design is not required) that outputs 1 if the combination is correct, 0 otherwise. This output is the COMB1 input to the system
2. the system waits for the user to enter the second combination and press the ENTER button. The combination is recognized by a combinational network (whose design is not required) that outputs 1 if the combination is correct, 0 otherwise. This output is the COMB2 input to the system
3. if both combinations are correct and have occurred in the correct sequence, the system unlocks the door (UNLOCK is set to 1) and waits to be reset. If any combination is wrong, the system doesn't unlock the door, it outputs 1 on the ERROR output and waits to be reset
4. reset is synchronous and brings the system back to the initial state.

Design a Moore FSM with 4 inputs RESET, COMB1, COMB2 and ENTER and 2 outputs UNLOCK and ERR exhibiting the above behavior. Assume the ENTER button when pressed generates a pulse for only one clock cycle.

1. Draw the state diagram (4 points)
2. Design the FSM resorting to the VHDL behavioural description style with 2 processes (4 points).
3. Design the FSM resorting in Verilog with 2 always blocks (4 points).

3. FSM-D design (16 points)

In **integer division** and **modulus**, the dividend is divided by the divisor into an integer quotient and a remainder. The integer quotient operation is referred to as integer division, and the integer remainder operation is the modulus. For example, if $X = 93$ and $Y = 17$, the integer quotient is 5 and the remainder is 8. A simple algorithm is based on repetitive subtraction. Design a FSM-D to perform integer division and modulus of pure binary numbers by repeated subtractions. Inputs X and Y are on 8 bits. Define a proper size for the integer quotient q and the remainder r . Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready` signal is asserted for one clock cycle. Reset is asynchronous. Values for outputs q and r are available at each computation step, but they make sense only when `ready` is 1.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (6 points)
2. Design a FSM-D (10 points):
 - create 2 processes for behavioural datapath description from the HLSM state diagram
 - create 2 processes for behavioural controller description from the HLSM state diagram.

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

no later than Saturday January 29 2 pm

- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.